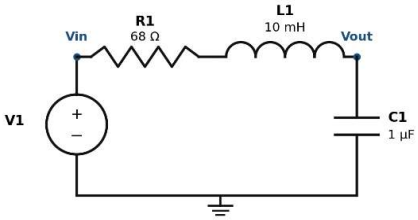
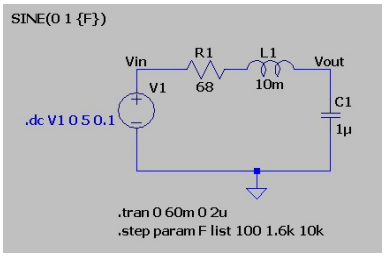
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5. Analyze the following second-order RLC low-pass network using sinusoidal transient excitation and Small-Signal AC Analysis.

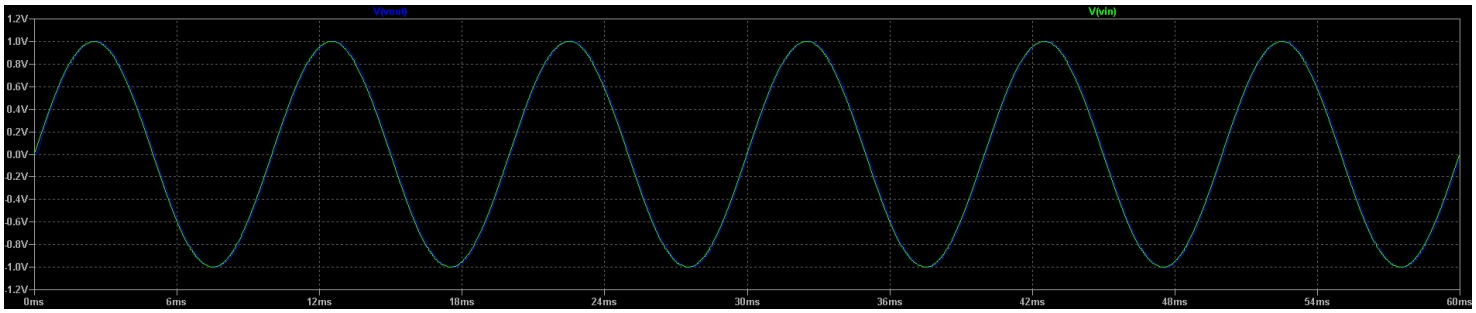




A. Sinusoidal transient response. Set V1 to the sine-wave source below and run a transient simulation. Plot V(vin) and V(vout), and measure the steady-state output amplitude and phase shift.

Source definition: SINE(0 1 {F})
Parameter sweep: .step param F list 100 1.6k 10k
Simulation directive: .tran 0 60m 0 2u
Compare the three frequency cases. If you prefer, the three frequencies may be simulated manually instead of using the .step directive.

100 Hz



Phase shift:

$Vin_{min_t} := 47.500031\ ms \qquad Vout_{min_t} := 47.568031\ ms$

$(100\ Hz \cdot (-360))\ deg \cdot (Vout_{min_t} - Vin_{min_t}) = -2.448\ deg$

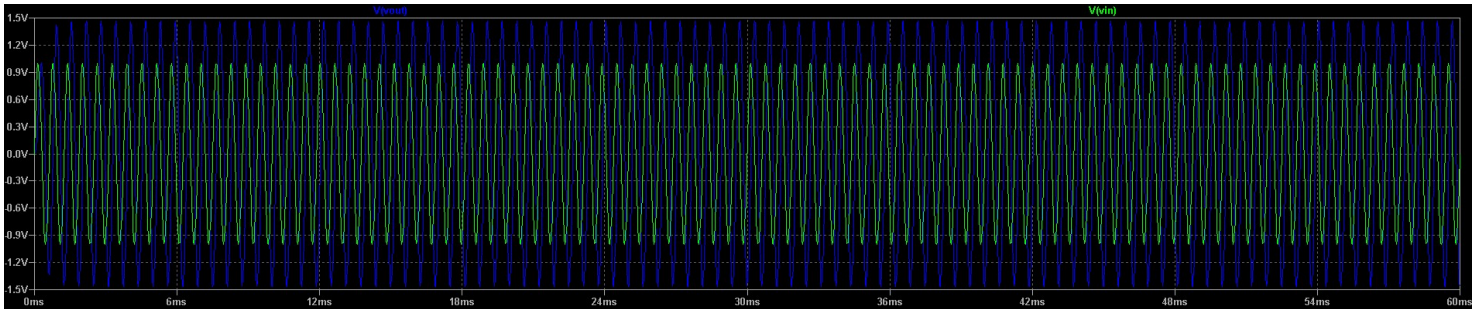
steady-state output amplitude

$Vout_{min} := -1.0030012\ volt$

$Vout_{max} := 1.0030012\ volt$

$$\frac{(Vout_{max} - Vout_{min})}{2} = 1.003\ V$$

1.6 kHz



Phase shift:

$Vin_{min_t} := 59.217908\ ms \qquad Vout_{min_t} := 59.375907\ ms$

$(1600\ Hz \cdot (-360))\ deg \cdot (Vout_{min_t} - Vin_{min_t}) = -91.0074\ deg$

steady-state output amplitude

$Vout_{min} := -1.462513\ volt$

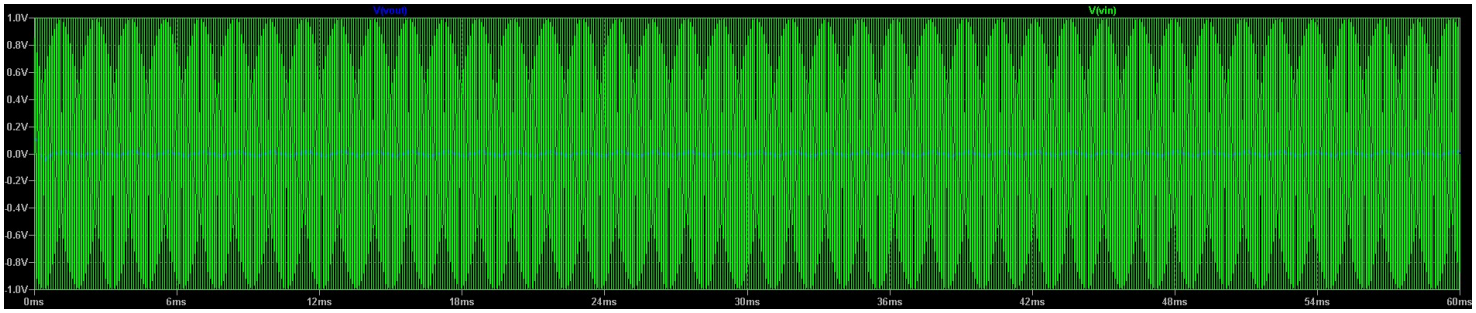
$Vout_{max} := 1.462513\ volt$

$$\frac{(Vout_{max} - Vout_{min})}{2} = 1.4625\ volt$$

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10 kHz



Phase shift:

$Vin_{min_t} := 59.274204\ ms \qquad Vout_{min_t} := 59.324204\ ms$

$(10000\ Hz \cdot (-360))\ deg \cdot (Vout_{min_t} - Vin_{min_t}) = -180\ deg$

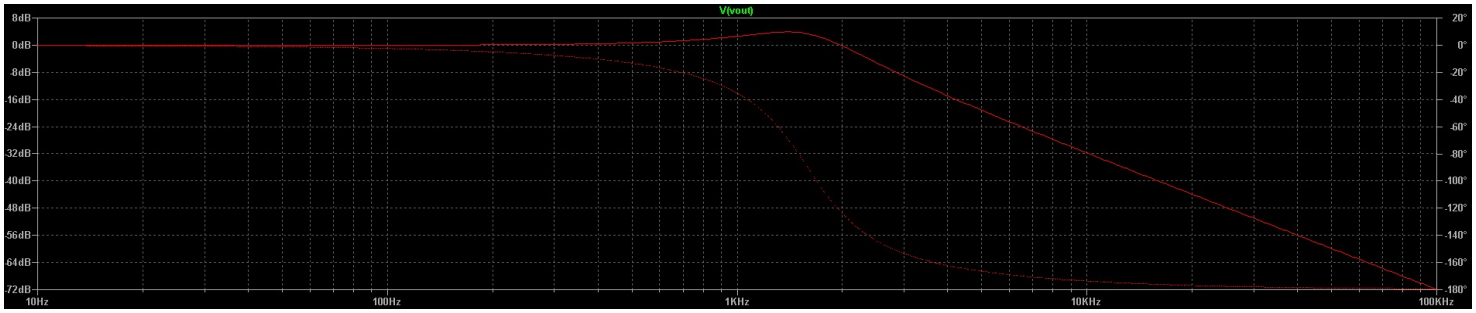
steady-state output amplitude

$Vout_{min} := -25.713556\ mV$

$Vout_{max} := 25.713556\ mV$

$$\frac{(Vout_{max} - Vout_{min})}{2} = 0.0257\ volt$$

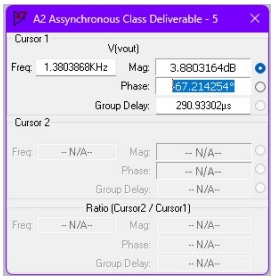
B. Small-Signal AC Analysis. Set the source Small signal AC amplitude to 1 V and run an AC sweep. Plot the magnitude in dB and phase of V(vout).
Simulation directive: .ac dec 100 10 100k
Determine the frequency of maximum output magnitude and compare it with the theoretical natural frequency f0 = 1/(2π√(L1·C1)). Also determine approximately where the response falls by 3 dB relative to its low-frequency gain.



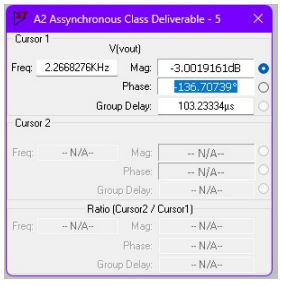
Theorical frequency of the maximum output magnitud

$L := 10\ mH \qquad C := 1\ \mu F$

$$f_0 := \frac{1}{2 \cdot \pi \cdot \sqrt{(L \cdot C)}} = 1.5915\ kHz$$



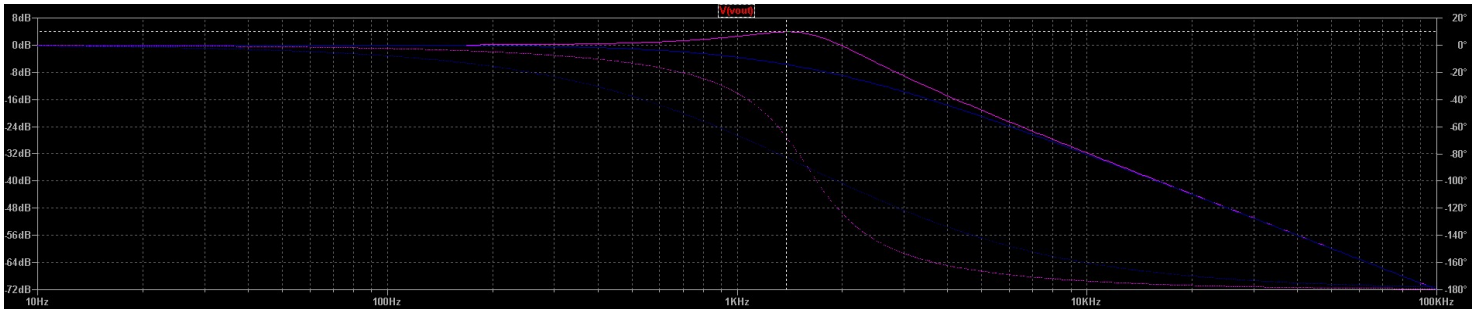
determine approximately where the response falls by 3 dB relative to its low-frequency gain.



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C. Damping comparison. Change R1 from 68 Ω to 220 Ω and repeat the AC analysis. Overlay or compare the two responses and explain how increasing resistance changes the resonance and damping of the circuit.
Important: The SINE definition is used for time-domain transient simulation. The AC amplitude field is used by Small-Signal AC Analysis. These are separate source settings in LTspice



Increasing R1 from 68 Ω to 220 Ω increases the damping because the resistor dissipates more energy, reducing the oscillatory exchange of energy between the inductor and capacitor. Consequently, the resonance peak disappears, and the magnitude decreases smoothly with increasing frequency.

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